UIL COMPUTER SCIENCE WRITTEN TEST

2025 REGION

APRIL 2025

General Directions (Please read carefully!)

1. DO NOT OPEN THE EXAM UNTIL TOLD TO DO SO.
2. There are 40 questions on this contest exam. You will have 45 minutes to complete this contest.
3. All answers must be legibly written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Clean erasures are necessary for accurate grading.
4. You may write on the test packet or any additional scratch paper provided by the contest director, but NOT on the answer sheet, which is reserved for answers only.
5. All questions have ONE and only ONE correct answer. There is a 2-point penalty for all incorrect answers.
6. Tests may not be turned in until 45 minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your test until told to do otherwise. You may use this time to check your answers.
7. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
8. All provided code segments are intended to be syntactically correct, unless otherwise stated. You may also assume that any undefined variables are defined as used.
9. A reference to many commonly used Java classes is provided with the test, and you may use this reference sheet during the contest. AFTER THE CONTEST BEGINS, you may detach the reference sheet from the test booklet if you wish.
10. Assume that any necessary import statements for standard Java SE packages and classes (e.g., `java.util`, `System`, etc.) are included in any programs or code segments that refer to methods from these classes and packages.
11. NO CALCULATORS of any kind may be used during this contest.

Scoring

1. Correct answers will receive **6 points**.
2. Incorrect answers will lose **2 points**.
3. Unanswered questions will neither receive nor lose any points.
4. In the event of a tie, the student with the highest percentage of attempted questions correct shall win the tie.

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

package java.lang

```
class Object
  boolean equals(Object anotherObject)
  String toString()
  int hashCode()

interface Comparable<T>
  int compareTo(T anotherObject)
    Returns a value < 0 if this is less than anotherObject.
    Returns a value = 0 if this is equal to anotherObject.
    Returns a value > 0 if this is greater than anotherObject.

class Integer implements Comparable<Integer>
  Integer(int value)
  int intValue()
  boolean equals(Object anotherObject)
  String toString()
  String toString(int i, int radix)
  int compareTo(Integer anotherInteger)
  static int parseInt(String s)

class Double implements Comparable<Double>
  Double(double value)
  double doubleValue()
  boolean equals(Object anotherObject)
  String toString()
  int compareTo(Double anotherDouble)
  static double parseDouble(String s)

class String implements Comparable<String>
  int compareTo(String anotherString)
  boolean equals(Object anotherObject)
  int length()
  String substring(int begin)
    Returns substring(begin, length()).
  String substring(int begin, int end)
    Returns the substring from index begin through index (end – 1).
  int indexOf(String str)
    Returns the index within this string of the first occurrence of str. Returns
    –1 if str is not found.
  int indexOf(String str, int fromIndex)
    Returns the index within this string of the first occurrence of str, starting
    the search at fromIndex. Returns –1 if str is not found.
  int indexOf(int ch)
  int indexOf(int ch, int fromIndex)
  char charAt(int index)
  String toLowerCase()
  String toUpperCase()
  String[] split(String regex)
  boolean matches(String regex)
  String replaceAll(String regex, String str)

class Character
  static boolean isDigit(char ch)
  static boolean isLetter(char ch)
  static boolean isLetterOrDigit(char ch)
  static boolean isLowerCase(char ch)
  static boolean isUpperCase(char ch)
  static char toUpperCase(char ch)
  static char toLowerCase(char ch)

class Math
  static int abs(int a)
  static double abs(double a)
  static double pow(double base, double exponent)
  static double sqrt(double a)
  static double ceil(double a)
  static double floor(double a)
  static double min(double a, double b)
  static double max(double a, double b)
  static int min(int a, int b)
  static int max(int a, int b)
  static long round(double a)
  static double random()
    Returns a double greater than or equal to 0.0 and less than 1.0.
```

package java.util

```
interface List<E>
class ArrayList<E> implements List<E>
  boolean add(E item)
  int size()
  Iterator<E> iterator()
  ListIterator<E> listIterator()
  E get(int index)
  E set(int index, E item)
  void add(int index, E item)
  E remove(int index)

class LinkedList<E> implements List<E>, Queue<E>
  void addFirst(E item)
  void addLast(E item)
  E getFirst()
  E getLast()
  E removeFirst()
  E removeLast()

class Stack<E>
  boolean isEmpty()
  E peek()
  E pop()
  E push(E item)

interface Queue<E>
class PriorityQueue<E>
  boolean add(E item)
  boolean isEmpty()
  E peek()
  E remove()

interface Set<E>
class HashSet<E> implements Set<E>
class TreeSet<E> implements Set<E>
  boolean add(E item)
  boolean contains(Object item)
  boolean remove(Object item)
  int size()
  Iterator<E> iterator()
  boolean addAll(Collection<? extends E> c)
  boolean removeAll(Collection<?> c)
  boolean retainAll(Collection<?> c)

interface Map<K,V>
class HashMap<K,V> implements Map<K,V>
class TreeMap<K,V> implements Map<K,V>
  Object put(K key, V value)
  V get(Object key)
  boolean containsKey(Object key)
  int size()
  Set<K> keySet()
  Set<Map.Entry<K, V>> entrySet()

interface Iterator<E>
  boolean hasNext()
  E next()
  void remove()

interface ListIterator<E> extends Iterator<E>
  void add(E item)
  void set(E item)

class Scanner
  Scanner(InputStream source)
  Scanner(String str)
  boolean hasNext()
  boolean hasNextInt()
  boolean hasNextDouble()
  String next()
  int nextInt()
  double nextDouble()
  String nextLine()
  Scanner useDelimiter(String regex)
```

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

Package java.util.function

```
Interface BiConsumer<T,U>
  void accept(T t, U u)

Interface BiFunction<T,U,R>
  R apply(T t, U u)

Interface BiPredicate<T,U>
  boolean test(T t, U u)

Interface Consumer<T>
  void accept(T t)

Interface Function<T,R>
  R apply(T t)

Interface Predicate<T>
  boolean test(T t)

Interface Supplier<T>
  T get()
```

UIL COMPUTER SCIENCE WRITTEN TEST – 2025 REGION

Note: Correct responses are based on **Java SE Development Kit 22 (JDK 22)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 22 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: `import static java.lang.System.*;`**

Question 1

Which of the following is equivalent to the expression 3948_{11} / 222_4?

A) 243_7 **B)** 72_8 **C)** $A2_{12}$ **D)** 1330_4 **E)** None are equivalent.

Question 2

What is output by the code to the right?

A) −2 **B)** 2 **C)** 0 **D)** −1

E) There is no output due to a compile error.

```
out.println(212 / 43 - 394 / 63);
```

Question 3

What is output by the code to the right?

A) true **B)** false **C)** TRUE **D)** FALSE

E) There is no output due to a runtime error.

```
out.printf("%B",-17);
```

Question 4

What is output by the code to the right?

A) nityStoit **B)** InfinityStone207

C) nitySto221 **D)** InfinityStonefi

E) There is no output due to a runtime error.

```
String s = "InfinityStone";
s.substring(1, s.length() - 2);
s += s.charAt(2) + s.charAt(5);
s.substring(3);
out.println(s);
```

Question 5

What is output by the code to the right?

A) true **B)** false

```
boolean a = true ^ true;
a = a & !a || a ^ !a && a | !a;
out.println(a);
```

Question 6

What is output by the code to the right?

A) 5 **B)** 6.0 **C)** 6

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
double g = 5.55;
int i = Math.round(g);
out.println(g);
```

Question 7

What is output by the code to the right?

A) 1211

B) 1311

C) 310

D) 1310

E) 210

```
int i = 9;
if(i < 10 || i++ > 10)
    out.print(1);
if(i < 10 && i++ > 10)
    out.print(2);
else
    out.print(3);
out.print(i);
```

Question 8

What is output by the code to the right?

A) 727 **B)** 81 **C)** 130 **D)** 87

E) There is no output due to a compile error.

```
int i = 78 % 13 ^ 28;
i += 212 - 39 | 51 / 2;
i = 134 & i * 5 + 77;
out.println(i);
```

Question 9

How many *'s are output by the code to the right?

A) 5 **B)** 6 **C)** 7 **D)** 8

E) There is no output due to an infinite loop.

```
for(int y=9; y < 123; y=y * 2 - 7)
        out.print("*");
```

Question 10

What is output by the code to the right?

A) [1, 118, 634, 212]

B) [33, 124, 74, 10, 3]

C) [0, 118, 634, 212]

D) [34, 122, 74, 10, 3]

E) There is no output due to a runtime error.

```
int[][] mat = new int[][] {
    {23, 5, -9, 212},
    {34, 89, 74, 10, 3},
    {341, 895, 284}
};
for(int i = 0; i < 3; i++) {
    mat[i][i] += mat[i][0]--;
    mat[0][i] -= mat[i][i]++;
    mat[0][i] *= -1;
}
out.println(
        Arrays.toString(mat[0]));
```

Question 11

What is output by the code to the right?

A) Koie

B) SB Th3

C) Koie 8293m

D) Th3

E) There is no output due to a runtime error.

```
String k =
    "123\nSB Th3\tKoie 8293m";
Scanner sc = new Scanner(k);
sc.nextLine();
sc.next();
sc.next();
out.println(sc.next());
```

Question 12

What is output by the code to the right?

A) 245

B) 185

C) 266

D) 336

E) There is no output due to an infinite loop.

```
int sum = 0;
for(int y = 5; y < 16; y++)
    for(int x = y; x >= 10; x--)
        sum += x;
out.println(sum);
```

Question 13

What is the order of precedence for the operators to the right?

A) II, I, III, IV **B)** II, III, IV, I

C) III, II, I, IV **D)** III, II, IV, I

E) II, III, I, IV

```
I. | (bitwise)
II. ++(pre)
III. *
IV. || (logical)
```

Question 14

What is output by the code to the right?

A) 80

B) 96

C) 64

D) 128

E) There is no output due to a runtime error.

```
int[] sizes = new int[] {
    Double.SIZE, Float.SIZE,
    Long.SIZE, Integer.SIZE,
    Short.SIZE, Byte.SIZE
};
Arrays.sort(sizes);
out.print(sizes[3]+sizes[4]);
```

Question 15

What is output by the line marked `//q15` in the code to the right?

A) `[GHIJK, LMNOP]`

B) `[GHIJK, LMNOP, WXYZ]`

C) `[ABC, DEF, QRS, TUV]`

D) `[ABC, DEF, QRS, TUV, WXYZ]`

E) There is no output due to an infinite loop.

```
ArrayList<String> a;
a = new ArrayList<String>();
a.add("ABC");a.add("DEF");
a.add("GHIJK");a.add("LMNOP");
a.add("QRS");a.add("TUV");
a.add("WXYZ");
a.removeIf(s -> s.length() < 4);
out.println(a); //q15
a.forEach(s ->{
     for (int i = 0; i < s.length(); i += 2)
          out.print(s.charAt(i));
}); //q16
```

Question 16

What is output by the line marked `//q16` in the code to the right?

A) `HJMOXZ`

B) `ACDFQSTV`

C) `BERU`

D) `GIKLNPWY`

E) There is no output due to a compile error.

Question 17

Which of the following lines in the code to the right will be the first to create a compile-time error?

A) `//line 01`

B) `//line 02`

C) `//line 03`

D) `//line 04`

E) `//line 05`

F) `//line 06`

G) `//line 07`

H) `//line 08`

I) `//line 09`

J) `//line 10`

K) The code will compile without error but will create a runtime error.

L) The code will both compile and run without error.

```
public void method1(
     int a,                         // line 01
     final int b                    // line 02
) {
     int c = a + b;                 // line 03
     final int d = a * b;           // line 04

     out.printf("%d %d %d %d\n",
          a, b, c, d);              // line 05

     a = a + 1;                     // line 06
     b = b + 1;                     // line 07
     c = c + 1;                     // line 08
     d = d + 1;                     // line 09

     out.printf("%d %d %d %d\n",
          a, b, c, d);              // line 10
}
```

Question 18

Which of the following lines in the code to the right will be the first to create a compile-time error?

A) `//line 01`

B) `//line 02`

C) `//line 03`

D) `//line 04`

E) `//line 05`

F) `//line 06`

G) `//line 07`

H) `//line 08`

I) `//line 09`

J) `//line 10`

K) `//line 11`

L) `//line 12`

M) `//line 13`

N) The code will compile without error but will create a runtime error.

O) The code will both compile and run without error.

```
public void method2(
     ArrayList<Integer> a,          // line 01
     final ArrayList<Integer> b     // line 02
) {
     ArrayList<Integer> c =
          new ArrayList<>(a);       // line 03
     final ArrayList<Integer> d =
          new ArrayList<>(a);       // line 04

     out.printf("%d %d %d %d\n",
          a.size(), b.size(),
          c.size(), d.size());      // line 05

     a.add(1);                      // line 06
     b.add(2);                      // line 07
     c.add(3);                      // line 08
     d.add(4);                      // line 09

     a = new ArrayList<>(a);        // line 10
     b = new ArrayList<>(b);        // line 11
     c = new ArrayList<>(c);        // line 12
     d = new ArrayList<>(d);        // line 13
}
```

Question 19

Which of the following concepts is demonstrated by the `add` methods in the classes to the right?

A) Overriding

B) Overloading

C) Abstraction

D) A and B

E) All of the above.

Question 20

Which of the following concepts is demonstrated by the `toString` method in the classes to the right?

A) Overriding

B) Overloading

C) Abstraction

D) A and B

E) All of the above.

Question 21

What is output by the line marked `//q21` in the code to the right?

A) `7`

B) `10`

C) `13`

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

Question 22

What is output by the line marked `//q22` in the code to the right?

A) `7`

B) `12`

C) `24`

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

Question 23

What is output by the line marked `//q23` in the client code to the right?

A) `AB`

B) `BA`

C) Output cannot be determined until runtime.

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
class A{

    private int y;
    String s;

    public A(String str) {
        s = str;
        y = 7;
    }

    public int get() {
        return y;
    }

    public void add() {
        y += 3;
    }

    public String toString() {
        return s;
    }

}

class B extends A{

    int y;

    public B(String str, int i) {
        super(str);
        y = i;
    }

    public void add(int i) {
        y += i;
    }

}
//////////client code//////////
A a = new A("A");
B b = new B("B", 12);
a.add();
out.println(a.get()); //q21
b.add(12);
out.println(b.get()); //q22
out.println(a+b); //q23
```

Question 24

Which of the following could replace **<1*>** in the code to the right and have the code compile without error (including the client code)?

A) `public` **B)** `final`

C) `protected` **D)** `static`

E) Replace all instances of **<1*>** with nothing (delete it and leave the space blank).

Question 25

Assuming that **<1*>** has been filled in properly in the code to the right, what is output by the client code to the right?

A)
```
Hello there!
May the force be with you.
```

B)
```
May the force be with you.
Hello there!
```

C)
```
Hello there!
Hello there!
```

D)
```
May the force be with you.
May the force be with you.
```

E) The output cannot be determined until runtime.

```
class Jedi {

    public static void speak() {
        out.println("May the force
            be with you.");
    }

    <1*> class ObiWan {

        public static void speak() {
            out.println("Hello
                there!");
        }

    }

}

////////// Client Code //////////
Jedi.ObiWan.speak();
Jedi.speak();
```

Question 26

Which of the following is equivalent to the logic circuit diagram to the right?

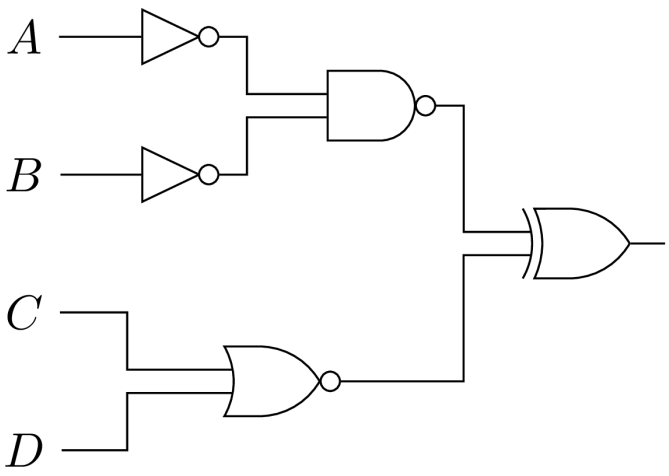
A) $\overline{\overline{A} + \overline{B}} \oplus \overline{C * D}$

B) $\overline{A} + \overline{B} \oplus \overline{C + D}$

C) $\overline{\overline{A} * \overline{B}} \oplus \overline{C + D}$

D) More than one of the above.

E) None of the above.



Question 27

How many permutations of `true` and `false` assignments to A, B, C, and D output `true` in the circuit diagram to the right?

A) 2 **B)** 4 **C)** 7 **D)** 10 **E)** 13

Question 28

Suppose we replaced input D with a second instance/pin/line of input C. How many permutations of `true` and `false` assignments to A, B, and C output `true` in the modified circuit diagram proposed?

A) 2 **B)** 4 **C)** 5 **D)** 7 **E)** 8

Question 29

What is the correct compile-time error with the code to the right as well as the appropriate way to resolve the error?

A) They keyword `final` is not a valid modifier to a class definition. To resolve, remove all `final` modifiers to the class definition.

B) Abstract classes can only define abstract methods and cannot define abstract fields. To resolve, remove the declaration of the `name` variable and the constructor from the abstract class and add the same variable definition to the three sub-classes. Then, instead of calling the parent constructor using `super`, instead assign the (now) local field `name` to the value previously passed using `super`.

C) Class definitions modified with the keyword `final` prevents other classes from inheriting from it. To resolve, remove the class definition of `EnemyJett`, as the client code does not rely on this class existing. Alternatively, if we needed the `EnemyJett` class to exist, have the `EnemyJett` class instead inherit from the `Agent` abstract class instead of the `Jett` class.

D) Classes that are declared as `abstract`, if inherited, need to use the `implements` keyword instead of the `extends` keyword. To resolve, replace all instances of `extends` with `implements`.

E) Abstract classes can never appear as the datatype of a variable declaration – only concrete classes or primitive datatypes can. To resolve, replace the datatype of the `tejo` variable with `Tejo`, and replace the datatype of the `jett` variable with the `Jett` object.

Question 30

Assuming that the correct compile-time error identified above has been resolved using its correct resolution method, what is output by the client code to the right?

A)
```
Tejo: This is how it ends!
Jett: Watch this!
```

B)
```
Tejo: This is how it ends!
Jett: Get out of my way!
```

C)
```
Tejo: This is how it ends!
Jett: Watch this!
Jett: Get out of my way!
```

D) There is no output, despite the program compiling and running without error.

E) The output cannot be determined until runtime.

```
abstract class Agent {
    public final String name;

    public Agent(String name) {
        this.name = name;
    }

    public abstract void ult();
}

final class Tejo extends Agent {
    public Tejo() {
        super("Tejo");
    }

    public void ult() {
        out.printf("%s: This is
            how it ends!\n",
            name);
    }
}

final class Jett extends Agent {
    public Jett() {
        super("Jett");
    }

    public void ult() {
        out.printf("%s: Watch
            this!\n", name);
    }
}

class EnemyJett extends Jett {
    public EnemyJett() {
        super("Jett");
    }

    public void ult() {
        out.printf("%s: Get out of
            my way!\n", name);
    }
}

////////// Client Code //////////
Agent tejo = new Tejo();
Agent jett = new Jett();
tejo.ult();
jett.ult();
```

Question 31

What can replace **<1*>** in the code to the right so that any object which implements the `Comparable` interface can be stored within the new object `DataStructure`?

A) `T`

B) `Comparable`

C) `? extends Comparable`

D) `T extends Comparable<T>`

E) None of the above

Question 32

Assuming that **<1*>** has been filled in properly, what can replace **<2*>** in the code to the right so that any object which implements the `Comparable` interface can be stored within the new object `DataStructure`?

A) `T`

B) `Comparable`

C) `? extends Comparable`

D) `T extends Comparable<T>`

E) None of the above

Question 33

What is the tightest upper bound on the time complexity of the `extract()` method in the code to the right?

A) $\mathcal{O}(n)$

B) $\mathcal{O}(n \log_2 n)$

C) $\mathcal{O}(\sqrt{n})$

D) $\mathcal{O}(1)$

E) $\mathcal{O}(\log_2 n)$

Question 34

The class `DataStructure` in the code to the right is an implementation of what well-known data structure?

A) Dequeue

B) AVL Tree

C) Binary Search Tree

D) Min-Heap

E) Max-Heap

```
class DataStructure<<1*>> {
    private ArrayList<<2*>> arr;

    public DataStructure(int n) {
       arr = new ArrayList<<2*>>();
    }

    private void swap(int i, int j) {
       <2*> temp = arr.get(i);
       arr.set(i, arr.get(j));
       arr.set(j, temp);
    }

    public void insert(<2*> val) {
       int i = arr.size();
       int parent = (i - 1) / 2;
       arr.add(val);

       while (i > 0 && arr.get(i).compareTo(
          arr.get(parent)) < 0)
       {
          swap(i, parent);
          i = parent;
          parent = (i - 1) / 2;
       }
    }

    public <2*> extract() {
       if (arr.isEmpty()) {
          return null;
       }

       <2*> min = arr.get(0);
       <2*> last = arr.remove(arr.size() - 1);
       if (!arr.isEmpty()) {
          arr.set(0, last);

          int i = 0;
          while (true) {
             int l = (2 * i) + 1;
             int r = (2 * i) + 2;
             int minIndex = i;

             if (l < arr.size() && arr.get(l)
                .compareTo(arr.get(minIndex))
                < 0)
             {
                minIndex = l;
             }
             if (r < arr.size() && arr.get(r)
                .compareTo(arr.get(minIndex))
                < 0)
             {
                minIndex = r;
             }

             if (minIndex == i) {
                break;
             }
             swap(i, minIndex);
             i = minIndex;
          }
       }
       return min;
    }
}
```

Question 35

Given a sorted array of $m = n^2$ elements, **in terms of n**, which of the following is the tightest upper bound for the fastest way to find the index of a specific element **among the m elements**?

A) $\mathcal{O}(m) = \boxed{\mathcal{O}(n^2)}$

B) $\mathcal{O}(m) = \mathcal{O}\left(\sqrt{n^2}\right) = \boxed{\mathcal{O}(n)}$

C) $\mathcal{O}(m) = \mathcal{O}((\log n^2)^2) = \mathcal{O}((2\log n)^2) = \mathcal{O}(4\log^2 n) = \boxed{\mathcal{O}(\log^2 n)}$

D) $\mathcal{O}(\mathrm{m}) = \mathcal{O}(\log(\mathrm{n}^2)) = \mathcal{O}(2\log \mathrm{n}) = \boxed{\mathcal{O}(\log \mathrm{n})}$

E) $\mathcal{O}(m) = \boxed{\mathcal{O}(1)}$

Question 36

Given a sorted singly-linked list of $m = n^2$ elements, **in terms of n**, which of the following is the tightest upper bound for the fastest way to find the index of a specific element **among the m elements**?

A) $\mathcal{O}(m) = \boxed{\mathcal{O}(n^2)}$

B) $\mathcal{O}(m) = \mathcal{O}\left(\sqrt{n^2}\right) = \boxed{\mathcal{O}(n)}$

C) $\mathcal{O}(m) = \mathcal{O}((\log n^2)^2) = \mathcal{O}((2\log n)^2) = \mathcal{O}(4\log^2 n) = \boxed{\mathcal{O}(\log^2 n)}$

D) $\mathcal{O}(\mathrm{m}) = \mathcal{O}(\log(\mathrm{n}^2)) = \mathcal{O}(2\log \mathrm{n}) = \boxed{\mathcal{O}(\log \mathrm{n})}$

E) $\mathcal{O}(m) = \boxed{\mathcal{O}(1)}$

Question 37

The code to the right demonstrates which of the following design patterns?

A) Singleton

B) Object Pool

C) Builder

D) Prototype

E) Factory Method.

```
class Something {
     private static Something something;

     private Something() {}
     public Something createSomething() {
         if(something == null) {
             return new Something();
         }
         return something;
     }
}
```

Question 38

Which of the following sorting algorithms **does not** have its worst case time complexity equal to $\mathcal{O}(n\log_2 n)$?

A) Heap Sort **B)** Merge Sort **C)** Quick Sort **D)** Option B and C. **E)** None of the above.

Question 39

Suppose that you have a complete, directed, weighted graph with no negative edges and you use Dijkstra's algorithm to solve the single-source shortest path problem. In terms of V, the number of vertices in the graph, and E, the number of edges in the graph, what is the tightest asymptotic upper bound for solving this problem? Express your answer in Big-$\mathcal{O}$ notation.

Question 40

What is the sum of all elements that remain in the stack after the process denoted to the right completes?

```
The stack is initially empty.
Push 18
Push 12
Push 25
Push 19
Pop
Pop
Push 17
Push 32
Push 36
Push 24
Peek
Peek
Pop
Push 56
Push 17
Pop
Process Complete
```